Advanced Dynare

Chapter II

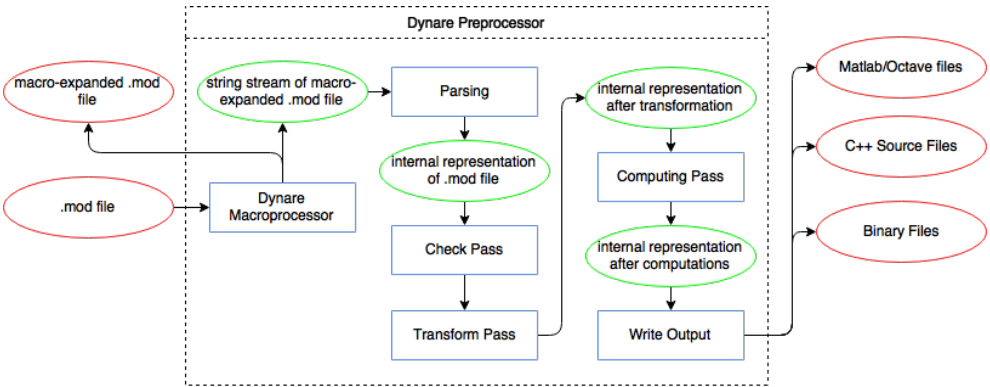
The Dynare preprocessor

Johannes Pfeifer

University of the Bundeswehr Munich

October 2025, Bank of Korea

Overview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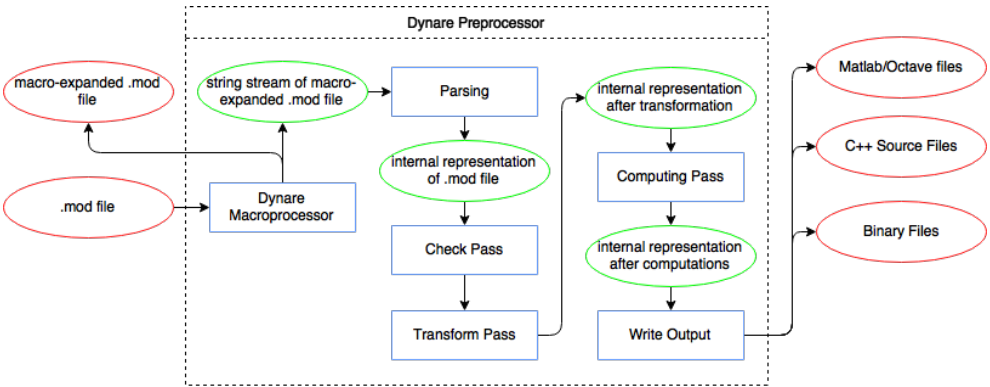
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Calling Dynare

- Dynare is called from the host language platform with the syntax

```
dynare FILENAME[.mod]
```

- This call can be followed by certain options:
  - Some of these options impact host language platform functionality
    → *e.g.* `nograph` prevents graphs from being displayed in MATLAB
  - Some cause differences in the output created by default
    → `notmpterms` prevents temporary terms from being written to the static/dynamic files
  - Others impact the functionality of the macroprocessor or the preprocessor
    → *e.g.* `nostrict` shuts off certain checks that the preprocessor does by default

```
dynare FILENAME[.mod] nostrict
```

- Matlab command line supports syntax completion since Dynare 6.x

Command line options in the mod-file

- Command line options can alternatively be defined in the first line of the .mod file
- Avoids to always have to invoke an option at the command line
  → particularly useful for the `nostrict` option during the development phase of a model
- Definition must be
  - a one-line Dynare comment, i.e. begin //
  - the options must be enclosed in `--+ options:` and `+--` and must be whitespace separated
  - As in the command line, if an option admits a value, the equal symbol must not be surrounded by spaces

```
// --+ options: json=compute, stochastic, nostrict +--
```

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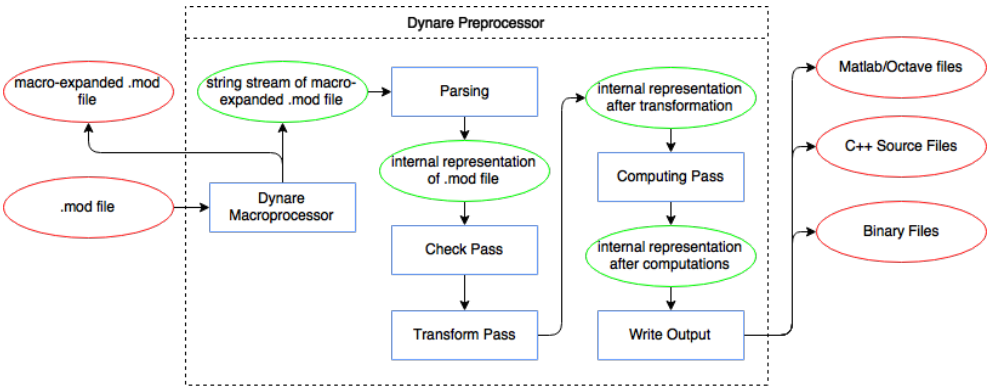
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Macro processing

- The Dynare macro language provides a set of macro commands that can be used in `mod` files
- The macro processor employs text expansions/inclusions to transform a `mod` file with macro commands into a `mod` file without macro commands
  → result can be stored using `savemacro` option
- The result is fed to the parser
  → use `onlymacro` to stop after macro processing before parsing

```
dynare FILENAME[.mod] onlymacro savemacro=macroresult
```

- Attention: the macro processor only does text substitution; objects computed in later steps are not available yet and cannot be conditioned on for that reason

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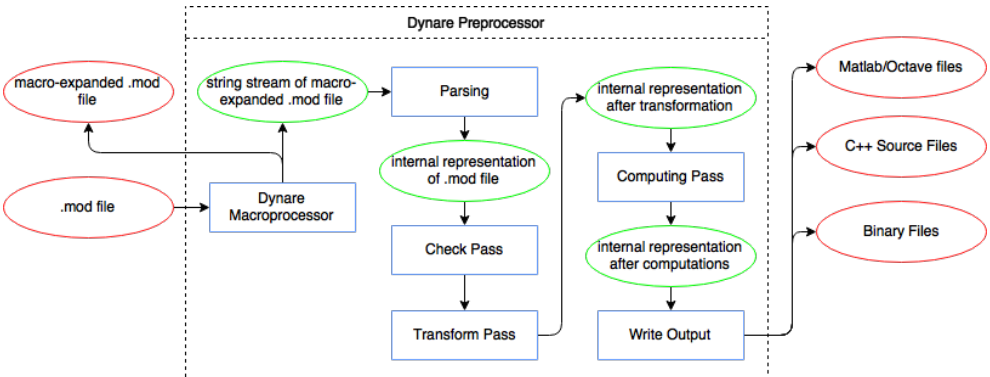
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Parsing overview

- Parsing is the action of transforming an input text (a `mod` file in our case) into a data structure suitable for computation
- The parser consists of three components:
  1. the lexical analyzer, which recognizes the "words" of the `mod` file (analog to the *vocabulary* of a language)
  2. the syntax analyzer, which recognizes the "sentences" of the `mod` file (analog to the *grammar* of a language)
  3. the parsing driver, which coordinates the whole process and constructs the data structure using the results of the lexical and syntax analyses

Undesired Parsing

- Dynare will try to parse all tokens it recognizes
- Code not recognized by the parser is directly passed to the preprocessor output
  $\rightarrow$ allows using MATLAB/Octave commands directly in the mod-file
- Causes problems when one wants to invoke commands containing recognized tokens

Bypassing the parsing

- The `verbatim;` block instructs Dynare not to parse text contained in it
- In the following example, `stoch_simul` would otherwise be recognized as a Dynare command during parsing

```
verbatim;
rhos = [ 0.8, 0.9, 1];
for iter = 1:length(rhos)
    set_param_value('rho',rhos(iter));
    [info, oo_, options_, M_] = stoch_simul(M_, options_, oo_, var_list_)
    if info(1)~=0
        error('Simulation failed for parameter draw')
    end
end
end;
```

1. Lexical analysis

- The lexical analyzer recognizes the "words" (or lexemes) of the language
- When invoked, the lexical analyzer reads the next characters of the input, tries to recognize a lexeme, and either produces an error or returns the associated token

2. Syntax analysis

- The `mod` file grammar is transformed into C++ source code by the program `bison`
- The grammar tells a story which looks like:
  - A `mod` file is a list of statements
  - A statement can be a `var` statement, a `varexo` statement, a `model` block, an `initval` block, ...
  - A `var` statement begins with the token `VAR`, then a list of `NAME`s, then a semicolon

Syntax analysis

- Using the list of tokens produced by lexical analysis, the syntax analyzer determines which "sentences" are valid in the language, according to a grammar composed of rules.
- `(1+3)*2; 4+5;` will pass the syntax analysis without error
- `1++2;` will fail the syntax analysis, even though it has passed the lexical analysis

3. Parsing driver

- The parsing driver opens the `mod` file and launches the lexical and syntactic analyzers
- It implements most of the semantic actions of the grammar
  → creates data structure representing the `mod` file
- If there is a parsing error
  - unknown keyword
  - syntax errors
  - use of undeclared symbols in model block, `initval` block...
  - use of a symbol incompatible with its type (*e.g.* parameter in `initval`, local variable used both in model and outside model)
  - ...

  it displays the line and column numbers where the first error occurred and exits
  → there may be others only displayed when the first error is fixed

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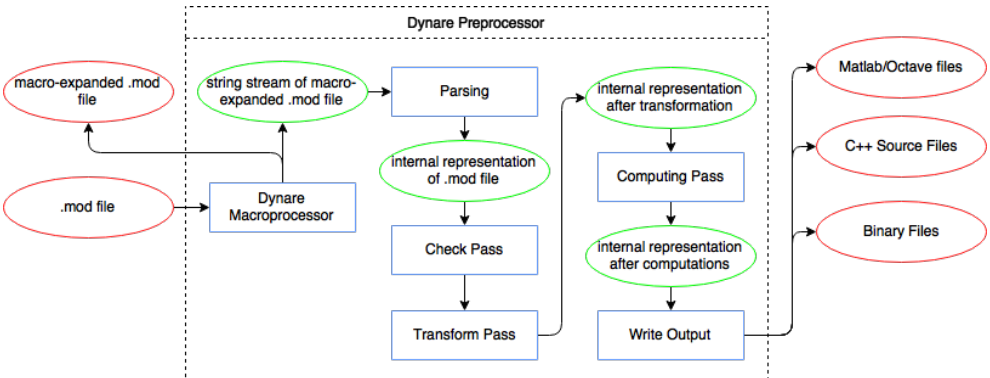
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Check pass

- Some errors in the `mod` file can be detected during parsing, but some other checks can only be done when parsing is completed...
- The check pass performs many checks. Examples include:
  - check there is at least one equation in the model (except if doing a standalone BVAR estimation)
  - checks for coherence in statements (*e.g.* options passed to statements do not conflict with each other, required options have been passed)
  - checks for coherence among statements (*e.g.* if `osr` statement is present, ensure `osr_params` and `optim_weights` or `planner_objective` statements are present (and not both))
  - checks for coherence between statements and attributes of `mod` file (*e.g.* `use_dll` is not used with `block` or `bytecode`)

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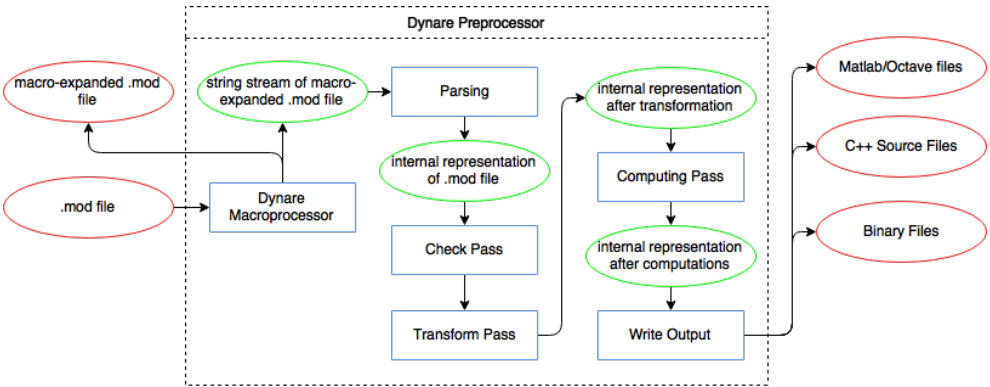
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Transform pass (1/2)

- The transform pass makes necessary transformations to prepare the `ModFile` object for the computing pass. Examples of transformations include:
  - creation of auxiliary variables and equations for leads, lags, expectation operator, differentiated forward variables, etc.
    → more information stored in `M_.aux_vars` and at Wiki
  - detrending of model equations if nonstationary variables are present
  - decreasing leads/lags of `predetermined_variables` by one period
  - addition of FOCs of Lagrangian for `ramsey_model`
  - setup of OccBin structure
  - addition of `dsge_prior_weight` initialization before all other statements if estimating a DSGE-VAR where the weight of the DSGE prior of the VAR is calibrated

Transform pass (2/2)

- Finally, checks are performed on the transformed model
- Examples include:
  - same number of endogenous variables as equations (not done in certain situations, *e.g.* `ramsey_model`, `discretionary policy`, where checks are done in Matlab)
  - correspondence among variables and statements, *e.g.* Ramsey, identification, perfect foresight solver, and perfect foresight solver are incompatible with `var_exo_det`
  - correspondence among statements, *e.g.* for DSGE-VAR with `bayesian_irf` option, the number of shocks must be equal to the number of observed variables

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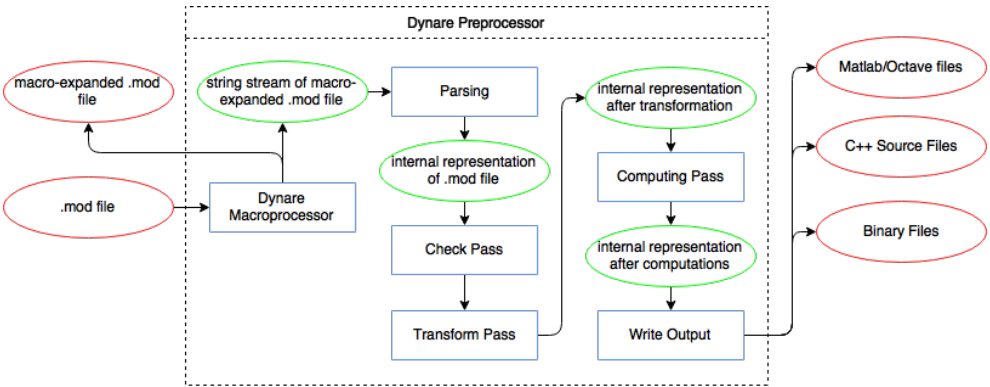
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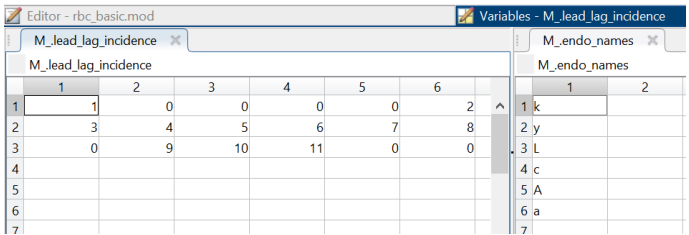
Overview of the computing pass

- Computing pass creates static model from dynamic one (by removing leads/lags)
- Determines which derivatives to compute and computes:
  - lead/lag variable incidence matrix
  - symbolic derivatives w.r.t. endogenous, exogenous, and parameters, if needed
  - equation normalization + block decomposition
  - temporary terms
- Asserts that equations declared linear are indeed linear (by checking that Hessian $== 0$)

Variables in the dynamic model

- `Dynamic Model` is characterized by leag/lag incidence matrix stored in `M_.lead_lag_incidence`:
  - `max_lag + max_lead + 1` rows (usually 3): the first row indicates $t-1$ (if applicable), the second row t, and the third row $t+1$ (if applicable)
  - one column for every endogenous symbol in order of declaration
    $\rightarrow$ includes endogenous and exogenous auxiliary variables created during the transform pass
  - elements of the matrix are either 0 (if the variable does not appear in the model) or correspond to the variable's column in the Jacobian of the dynamic model



Static versus dynamic model

- The static model is simply the dynamic model without leads and lags, used to characterize the steady state
- The Jacobian of the static model is used in the (MATLAB) solver for determining the steady state

Example

- suppose dynamic model is $2x_t \cdot x_{t-1} = 0$
- static model is $2x^2 = 0$, whose derivative w.r.t. x is $4x$
- dynamic derivative w.r.t. x_t is $2x_{t-1}$, and w.r.t. x_{t-1} is $2x_t$
- removing leads/lags from dynamic derivatives and summing over the two partial derivatives w.r.t. x_t and x_{t-1} gives $4x$

Which derivatives to compute?

- In deterministic mode:
  - static Jacobian w.r.t. endogenous variables only
  - dynamic Jacobian w.r.t. endogenous variables only
- In stochastic mode:
  - static Jacobian w.r.t. endogenous variables only
  - dynamic Jacobian w.r.t. endogenous, exogenous, and deterministic exogenous variables
  - dynamic Hessian w.r.t. endogenous, exogenous, and deterministic exogenous variables
  - possibly dynamic 3rd derivatives (if `order` option ≥ 3)
  - possibly dynamic Jacobian and/or Hessian w.r.t. parameters (if `identification` or `analytic_` needed for `estimation` and `params_derivs_order` > 0)
- For Ramsey policy: the same as above, but with one further order of derivation than declared by the user with `order` option

Temporary terms (1/2)

- When the preprocessor writes equations and derivatives in its outputs, it takes advantage of sub-expression sharing
- In MATLAB static and dynamic output files, equations are preceded by a list of **temporary terms**
- These terms are variables containing expressions shared by several equations or derivatives
  → greatly enhances the computing speed of the model residual, Jacobian, Hessian, or third derivative

Original:

```
residual(0)=x+y^2-z^3;
residual(1)=3*(x+y^2)+1;
```

Optimized:

```
T1=x+y^2;
residual(0)=T1-z^3;
residual(1)=3*T1+1;
```

Temporary terms (2/2)

- Expression storage in the preprocessor implements maximal sharing, but this is not optimal for the MATLAB output files, because creating a temporary variable also has a cost (in terms of CPU and of memory)
- Computation of temporary terms implements a trade-off between:
  - cost of duplicating sub-expressions
  - cost of creating new variables
- Algorithm uses recursive cost calculation that marks some nodes as being "temporary"
- *Problem*: redundant with optimizations done by the C/C++ compiler (when Dynare is in DLL mode)
  $\Rightarrow$ compilation very slow on big models

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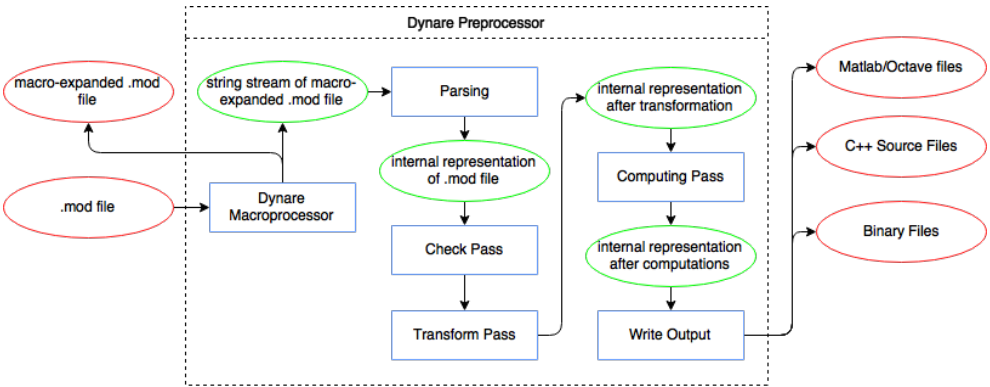
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Output overview

- If `mod` file is `mymodel.mod`, all created files will be in a namespace `mymodel` (`+mymodel` subfolder)
- Main output file is `mymodel.driver`, containing:
  - general initialization commands
  - symbol table output (variable and parameters names etc.)
  - lead/lag incidence matrix
  - call to MATLAB functions corresponding to the statements of the `mod` file
- Subsidiary output files:
  - for the static model (residuals, temporary terms, derivatives)
  - for the dynamic model (residuals, temporary terms, derivatives)
  - one for the auxiliary variables in the dynamic model (if relevant)
  - one for the steady state file (if relevant)
  - for the planner objective and Lagrange multipliers (static residuals and derivatives, if relevant)
  - one each for the static and dynamic parameter derivatives (if required)